

Project Name	RTDIP Timeseries Forecasting
Online team meeting	https://fau.zoom-x.de/j/65502405196?pwd=8H73lyixQfqKrnO8Eb47kawnuKCnGp.1
Production system (if any)	
Test system (if any)	
GitHub repository	https://github.com/amosproj/amos2025ws03-rtdip-timeseries-forecasting
GitHub feature board	https://github.com/orgs/amosproj/projects/91/
GitHub imp-squared backlog	https://github.com/orgs/amosproj/projects/96
Team T-shirt (white)	https://www.shirtinator.de/s/QWcDXTKGS72ISnrRNZq1hg
Team T-shirt (black)	https://www.shirtinator.de/s/0yx3duRjSzW-5hP32KmZ1w
Additional materials	https://discord.gg/KJXGmjcs
Team mailing list	oss-amos-projX@lists.fau.de
Happines Index App	https://happy-amos.appspot.com/Courses
Planning Poker	https://planningpokeronline.com/DMhnF5cOAK9ffF5jsRf2/

Last Name	First Name	GitHub User Name	Email Address
Böhm	Luca	qw3rat	luca.boehm@fau.de
Meusling	Patrick	SirGankalot	meusling@campus.tu-berlin.de
Arifin	Hafidz	zenzeii	h.arifin@campus.tu-berlin.de
Huy	Christoph	Perimora	christoph.huy@campus.tu-berlin.de
Selbig	Simon	simonselbig	simon.selbig@gmx.de
Haseeb	Abdul	abdulhaseeb-se	abdul.ah.haseeb@fau.de
Pohnke	Hannes	hpfref	hannes.j.pohnke@gmail.com
Jokiel	Rene	BelmontR	rene.jokiel@fau.de
Khabouze	Mehdi	Mehdi-kbz	khabouze@campus.tu-berlin.de

#	Meeting Day	Product Owner		Software Developer	Release Manager	Scrum Master	Comment
		Review	Planning				
1	2025-10-15	Patrick Meusling	Hafidz Arifin	Everyone else	N/A	Luca	
2	2025-10-22	Hafidz Arifin	Patrick Meusling	Everyone else	Christoph	Luca	
3	2025-10-29	Patrick Meusling	Hafidz Arifin	Everyone else	Mehdi Khabouze	Luca	
4	2025-11-05	Hafidz Arifin	Patrick Meusling	Everyone else	Rene Jokiel	Luca	
5	2025-11-12	Patrick Meusling	Hafidz Arifin	Everyone else	Abdul Haseeb	Luca	
6	2025-11-19	Hafidz Arifin	Patrick Meusling	Everyone else	Hannes Pohnke	Luca	
7	2025-11-26	Patrick Meusling	Hafidz Arifin	Everyone else	Simon Selbig	Luca	Mid-term due
8	2025-12-03	Hafidz Arifin	Patrick Meusling	Everyone else	Rene Jokiel	Luca	
9	2025-12-10	Patrick Meusling	Hafidz Arifin	Everyone else	Mehdi Khabouze	Luca	
10	2025-12-17	Hafidz Arifin	Patrick Meusling	Everyone else	Christoph Huy	Luca	
11	2026-01-07	Patrick Meusling	Hafidz Arifin	Everyone else	Hannes Pohnke	Luca	
12	2026-01-14	Hafidz Arifin	Patrick Meusling	Everyone else	Simon Selbig	Luca	
13	2026-01-21	Patrick Meusling	Hafidz Arifin	Everyone else	Mehdi Khabouze	Luca	
14	2026-01-28	Hafidz Arifin	Patrick Meusling	Everyone else	Hannes Pohnke	Luca	
15	2026-02-04	Patrick Meusling	Hafidz Arifin	Everyone else	Christoph Huy	Luca	Demo day!
	2026-02-11	Hafidz Arifin	Patrick Meusling	Everyone else		Luca	Retrospective
Product owners, software developers, and Scrum Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here							

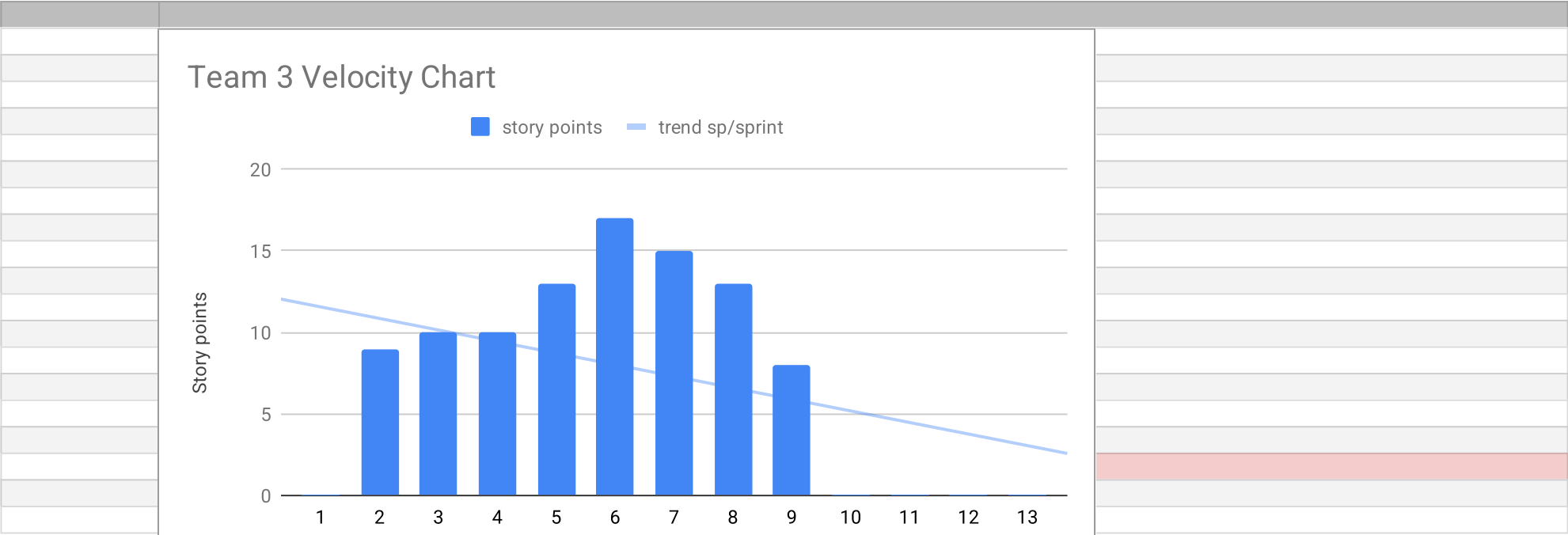
Goals	Aquire new skills
	Produce a functioning and valuable product (a 1.0)
Meeting norms	We respect the opinions of others and we show up on time
	Summary of Partner Meeting required
Working norms	Clean, testable code with clear commit messages, code comments
	We respect other people's work
Coordination norms	Seperate branches for issues, second person review
	We balance workload among the team
Communication norms	Daily check the corresponding channels (discord)
	We communicate constructively, make sure to communicate possible problems (absence, etc.)
Consideration norms	We discuss issues openly
	We vote in case we can't reach a consensus
Cont. improvement norms	We encourage critique and improvement efforts
	Encourage reaching out for help (use the strength/skills of others)
Rewards	We treat ourselves to a sweet of choice for good work
Sanctions	Push Ups (amount decided on the case)
Signatures	
Scrum Master	Luca Böhm
Product owner	Patrick Meusling
Product owner	Hafidz Arifin
Software developer	Abdul Haseeb
Software developer	Rene Jokiel
Software developer	Simon Selbig
Software developer	Hannes Pohnke
Software developer	Christoph Huy
Software developer	Mehdi Khabouze

Product Vision	Project Mission
To empower organizations with open-source, scalable, and transparent forecasting capabilities that enable data-driven decision-making across industries. By extending RTDIP with advanced forecasting and anomaly detection components as well as sector specific datasets, we envision a future where time series insights such as trends, seasonality, and predictive patterns are seamlessly integrated into real-time data pipelines. This will allow businesses to understand historical data, anticipate future behavior, optimize operations, and improve efficiency using accessible, community-driven, and production-ready forecasting tools.	To design, develop, and contribute open-source forecasting components for the RTDIP platform that enable trend analysis, anomaly detection, and predictive modeling on time series data. Our mission within this project is to research and implement forecasting techniques using Python and Apache Spark, validate and enrich them with real-world datasets, and ensure they meet RTDIP's modular, tested, and well-documented standards. By doing so, we will enhance RTDIP's functionality and provide the open-source community and industry users such as Shell with reliable, scalable forecasting capabilities that integrate seamlessly into existing data pipelines.

Term	Definition
RTDIP (Real-Time Data Ingestion Platform)	An open-source project under the Linux Foundation designed to simplify the ingestion, transformation, and storage of data from various sources using scalable cloud technologies.
Forecasting	The process of predicting future values or trends based on historical time series data.
Time Series Data	Data collected over time at regular intervals, often used to analyze patterns such as trends and seasonality.
Apache Spark	An open-source distributed computing system that provides fast data processing for large-scale datasets.
PySpark	The Python API for Apache Spark, allowing developers to write Spark applications using Python.
Delta Lake	A storage layer that brings reliability, consistency, and performance to data lakes by supporting ACID transactions and schema enforcement.
Anomaly Detection	The identification of unusual patterns or outliers in data that do not conform to expected behavior.
Predictive Modeling	A statistical or machine learning approach that uses historical data to predict future outcomes.
Time Series Decomposition	The process of breaking down a time series into its core components such as trend, seasonality, and residuals.
ETL (Extract, Transform, Load)	A data pipeline process that extracts data from sources, transforms it into a suitable format, and loads it into a storage system.
Open Source	Software that is freely available to use, modify, and distribute, typically developed collaboratively by a community.
Linux Foundation	A nonprofit organization that supports open-source software projects and fosters collaboration across industries.
EasyCLA (Easy Contributor License Agreement)	A system used by the Linux Foundation to manage contributor license agreements, ensuring legal compliance for open-source contributions.
Modular Architecture	A software design principle that divides a system into separate, interchangeable components that can be developed and maintained independently.
Unit Testing	The practice of testing individual units or components of software to ensure they work as intended.
Documentation	Written descriptions and guides that explain the design, usage, and functionality of a software system.
Data Pipeline	A set of processes that move, transform, and store data from one system to another in a structured and automated way.
Scalability	The ability of a system to handle increasing amounts of work or data by adding resources.
Contribution Guidelines	A set of rules and best practices that contributors must follow to ensure consistency and quality in open-source projects.

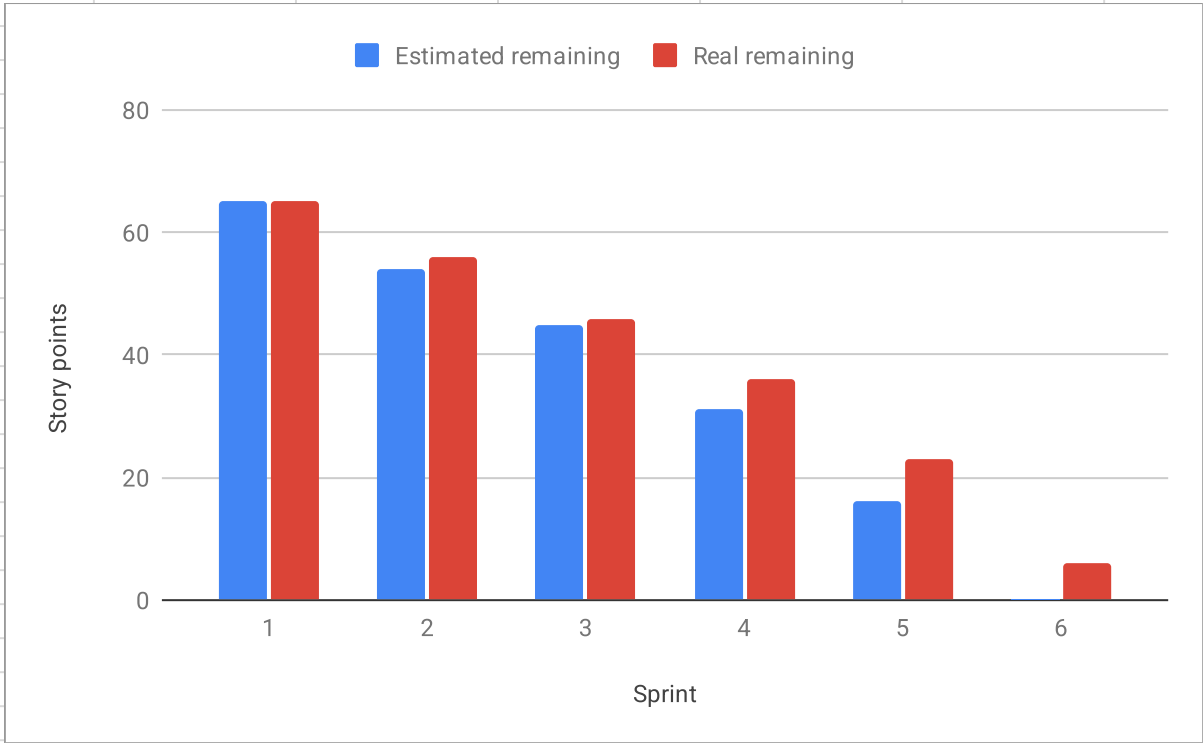
Sprint #	Sprint goal
1	None
2	None
3	None
4	Optional
5	Establish a working foundation for the forecasting pipeline
6	Integrate the pipeline into the existing software
7	Advance the pipeline
8	Add more Datasets to the pipeline
9	Testing and Analysing the new Datasets
10	Adding new and different models to the pipeline
11	Testing and Analysing the new Models
12	Finalize the Product
13	Finalize for Demo day
14	Merging the product with the partner, coordinating the merge (they said this might take a bit of time)
15	Backup Sprint

Sprint #	Story Points Realized
1	0
2	9
3	10
4	10
5	13
6	17
7	15
8	13
9	8
10	0
11	0
12	0
13	0
14	0
	PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB



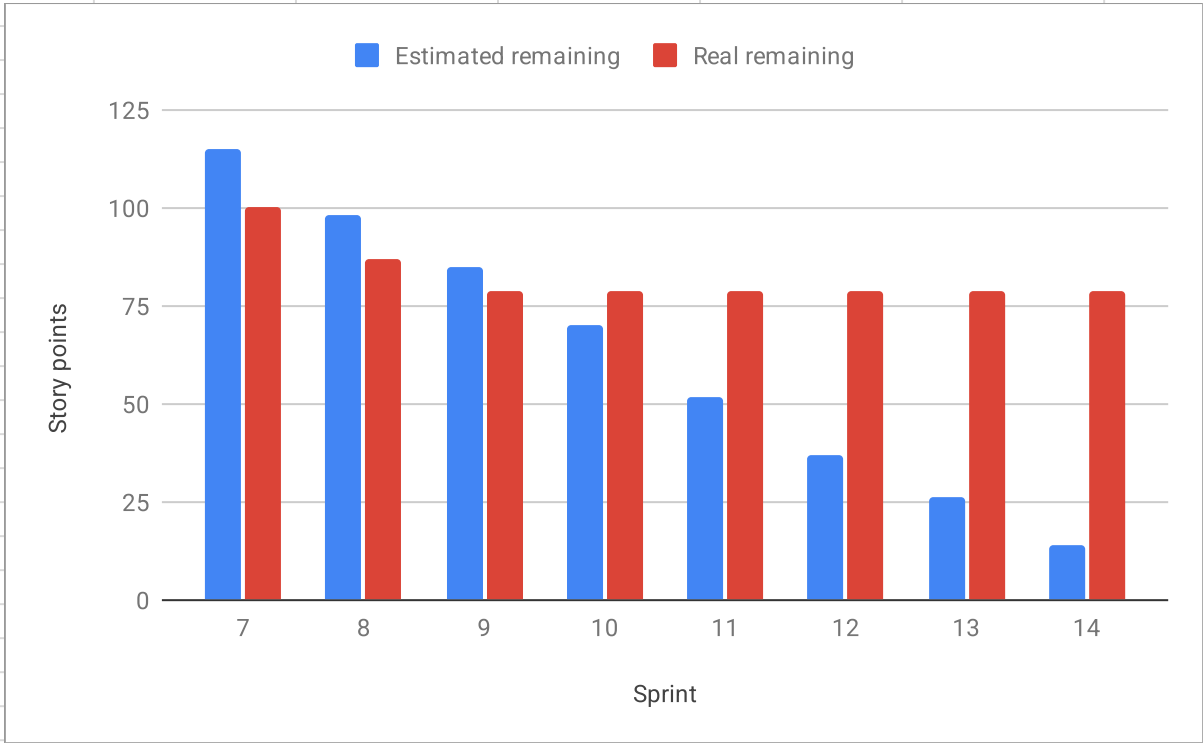
Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			65	65		
Sprints						
1	None		0	65	0	65
2	None		11	54	9	56
3	None		9	45	10	46
4	Optional		14	31	10	36
5	Establish a working foundation for the forecasting pipeline		15	16	13	23
6	Integrate the pipeline into the existing software		16	0	17	6
Features						
1	None					
2	None					
		Run the software on your own machine	1		1	
		Understanding the rtdip software	2		2	
		Software Architecture	2		2	
		Find open source time series datasets	2		2	
		Initialize software bill of materials	2		2	
		Research about time series forecasting fundamentals	2		NotCompleted	
3	None					
		Research Possible Models Useful for Shell Dataset	2		2	
		Update our Forked Repository with Latest RTDIP Release	1		1	
		Prepare and Demo Project Build Process	1		2	
		Perform Data Exploration on Shell Dataset	3		3	
		Research about time series forecasting fundamentals	2		2	
4	Optional					
		Getting to Know HPC of AMOS	2		NotCompleted	
		Perform Data Exploration on Opensource Dataset 1	2		3	
		Preprocess Shell Dataset	3		3	
		Implementing a basic Model pipeline	2		2	
		Research Possible Models Useful fo Opensource Dataset	2		NotCompleted	
		CI/CD Pipeline fail bug	3		2	
5	Establish a working foundation for the forecasting pipeline					
		Choosing Models, Datasets, Error Measure Metrics	2		2	
		Create Build Process Video	2		2	
		Model Training and Evaluation with Shell Dataset	3		3	
		Visualization of Results	2		1	
		Preprocess Opensource Dataset 1	3		3	
		PO Homework	3		2	

[illegible]



Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			115	115		
Sprints						
7	Advance the pipeline		17	115	15	100
8	Add more Datasets to the pipeline		13	98	13	87
9	Different Methods to Improve Forecast Accuracy		15	85	8	79
10	Customer Feedback and Adjustments		18	70	0	79
11	Improving Forecast Accuracy		15	52	0	79
12	Smaller features for more/different preprocessing etc.		11	37	0	79
13	Demo Day Preparation		12	26	0	79
14	Finish the Project		14	14	0	79
Features						
7	Advance the pipeline					
		Collect and Document Customer Feedback on Project Progress	1		1	
		Adjust and Fix CI/CD Pipeline	3		Not completed	
		Model Optimization and Improvement	3		3	
		Implementation of Time Series Decomposition	3		3	
		Implementation of Anomaly Detection	5		5	
		Model Training and Evaluation of Dataset SCADA	2		3	
8	Add more Datasets to the pipeline					
		Perform Data Exploration additional Opensource Dataset	3		3	
		Implement RTDIP Ingestion	2		2	
		Restructure and Refactor Wiki	2		2	
		Standardize Visualization	3		3	
		Model Optimization and Improvement for Timeseries Forecasting	3		3	
9	Different Methods to Improve Forecast Accuracy					
		Optimization and Improvement of Anomaly Detection Approach	3		3	
		Timeseries Forecasting Improvement with Other Models (Person 1)	3		3	
		Integrate Timeseries Forecasting Results Visualization into the SDK	3		Not Completed	
		Apply Timeseries Decomposition on shell Dataset	3		2	
		Timeseries Forecasting Improvement with Other Models (Person 2)	3		Not Completed	
10	Customer Feedback and Adjustments					
		Refactor Components for Spark-Compatible API	3			
		Document previous features in the software architecture	3			
		Adjusting previous features to base environment	2			
		Base Environment	8			
		Add Visualization Component for Anomaly Detection	2			

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
11	Improving Forecast Accuracy					
		Testing our own and new model approaches	3			
		Timeseries Forecasting Improvement by Combining FEs from Best Models	3			
		Timeseries Forecasting Improvement with Exogenous Variables	3			
		Timeseries Forecasting Improvement with Other Models	3			
		Timeseries Forecasting Improvement by Trying Different Versions of Models	3			
12	Finish the Product					
		Adjusting our new approaches based on Customer Feedback	3			
		Final Refactoring	2			
		Testing our own and new preprocessing approaches	3			
		Testing our own and new postprocessing approaches	3			
13	Demo Day Preparation					
		Create demo day slide	3			
		Create demo video	3			
		Finalize user, (technical) design, and build/deploy documentation	3			
		Final Refactoring and Cleaning Up Repository	3			
14	Finish the Project					
		Create Project Report	3			
		Tasks related to Industry Partner	5			
		Merging the code with main repo of partner	3			
		Final Bug Fixes	3			
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				



#	Feature Definition of Done	Sprint Release Definition of Done	Project Release Definition of Done
	Acceptance criteria are met.		
	Work are pushed to the Github repository.		
	Create a branch for each backlog items (coding)		
	A pull request is created for each related branch.		
	The work products in the pull requests are reviewed.		
	The corresponding branches are merged and closed.		
	The bill of materials section of the planning documents is updated.		
	Work needs to be documented in the corresponding wiki section		
	For new features unit test have to be written.		
	Update our forked repository with the latest RTDIP release.		
	If the task involves coding, the implementation is integrated into the RTDIP framework and verified to function correctly within it.		
	All unit tests must pass successfully in the continuous integration (CI) pipeline.		
	All assignees must communicate and coordinate with each other to complete the task.		
		Release candidates with a working and meaningful update to the previous sprint is tagged.	
		Previously established features must continue to work.	
			The RTDIP forecasting components can be successfully built and deployed within the RTDIP environment.
			All automated tests and validation checks for the developed components pass without errors.
			The implemented forecasting features have been tested end-to-end and verified through a basic user workflow.
			Developer documentation is complete, clearly describing the architecture, setup, and contribution process for RTDIP.
			User documentation (usage examples, configuration steps, expected outputs) is finalized and up to date.
			The final release has been reviewed and approved by all project team members and the industry partner (Shell).

Type	Link / reference
User Documentation	https://github.com/amosproj/amos2025ws03-rtdip-timeseries-forecasting/wiki/Documentation#user-documentation
Design Documentation	https://github.com/amosproj/amos2025ws03-rtdip-timeseries-forecasting/wiki/Documentation#design-documentation
Build & Deploy Documentation	https://github.com/amosproj/amos2025ws03-rtdip-timeseries-forecasting/wiki/Documentation#build-and-deploy-documentation

System Components					Third-Party Dependencies and Libraries					Tools				
Name	Category	Status	Description		Name	Category	Version	License	Description	Name	Category	Version	Description	
Frontend	UI Component	Active	ReactJS - Frontend with TypeScript		ReactJS	Library	18.2.0	MIT	Open-source JavaScript library for building user interfaces	VS Code	Development	1.85.0	Open-source code editor with integrated support for Git, Docker, and remote Explorer	
Backend	API Service	Active	NodeJS - Backend with ExpressJS		ExpressJS	Library	4.18.2	MIT	Minimalist JavaScript web framework for NodeJS	Git	Development	2.43.0	Distributed version control system for tracking changes, managing branching, and merging	
Database	Storage	Active	MongoDB - NoSQL database		MongoDB	Database	6.0.14	AGPL	Open-source NoSQL database with flexible, JSON-like documents	Docker	Development	24.0.6	Open-source container management platform for automating application deployment	
Cache	Cache Service	Active	Redis - In-memory data structure store		Redis	Cache	7.0.15	BSD	Open-source in-memory data structure store for caching, messaging, and real-time analytics	Webpack	Development	5.90.0	Open-source module bundler for JavaScript applications	
Queue	Queue Service	Active	RabbitMQ - Message broker		RabbitMQ	Queue	3.13.1	Apache License	Open-source message broker implementing the AMQP protocol	ESLint	Development	8.56.0	Open-source JavaScript linting tool for identifying and reporting on code quality issues	
Search	Search Service	Active	Elasticsearch - Search engine		Elasticsearch	Search	8.14.0	Apache License	Open-source search engine for full-text and structured data	Prettier	Development	3.2.5	Open-source code formatter for JavaScript, CSS, and HTML	
Logging	Logging Service	Active	Logstash - Log processing pipeline		Logstash	Logging	8.14.0	Apache License	Open-source log processing pipeline for collecting, parsing, and forwarding log data	Stylelint	Development	15.11.0	Open-source CSS linter and formatter	
Monitoring	Monitoring Service	Active	Prometheus - Monitoring system		Prometheus	Monitoring	2.47.0	Apache License	Open-source monitoring system for gathering metrics and alerts	Autoprefixer	Development	10.4.19	Open-source CSS autoprefixer for adding vendor prefixes	
Deployment	Deployment Service	Active	Kubernetes - Container orchestration		Kubernetes	Deployment	1.29.0	Apache License	Open-source container orchestration system for automating deployment and scaling	Canary	Development	1.16.0	Open-source tool for managing and automating deployment	
CI/CD	CI/CD Pipeline	Active	Jenkins - Continuous integration		Jenkins	CI/CD	2.421	MIT	Open-source automation server for implementing CI/CD pipelines	Netlify	Development	3.10.0	Open-source platform for static site hosting and serverless functions	
Testing	Testing Framework	Active	Jest - JavaScript testing framework		Jest	Testing	29.7.0	MIT	Open-source JavaScript testing framework for unit and integration tests	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Security	Security Framework	Active	OWASP ZAP - Web application security scanner		OWASP ZAP	Security	2.10.0	GPL	Open-source web application security scanner for identifying vulnerabilities	SendGrid	Development	7.13.0	Open-source email delivery service for sending transactional and marketing emails	
Payment	Payment Gateway	Active	Stripe - Payment processing		Stripe	Payment	10.12.0	Proprietary	Open-source payment processing API for accepting online payments	Algolia	Development	4.19.0	Open-source search-as-a-service for fast and relevant search results	
Analytics	Analytics Service	Active	Google Analytics - Web analytics		Google Analytics	Analytics	4.17.0	Proprietary	Open-source web analytics service for tracking and reporting website usage	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Localization	Localization Service	Active	i18next - Internationalization framework		i18next	Localization	22.1.6	MIT	Open-source internationalization framework for web and mobile applications	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Reporting	Reporting Service	Active	Tableau - Business intelligence		Tableau	Reporting	2023.3	Proprietary	Open-source business intelligence tool for data visualization and analysis	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Compliance	Compliance Framework	Active	GDPR - General Data Protection Regulation		GDPR	Compliance	1.0.0	Proprietary	Open-source framework for ensuring compliance with GDPR and other data privacy regulations	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Accessibility	Accessibility Framework	Active	WCAG - Web Content Accessibility Guidelines		WCAG	Accessibility	2.1.4	Proprietary	Open-source guidelines for making web content more accessible to people with disabilities	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Performance	Performance Framework	Active	Lighthouse - Performance auditing		Lighthouse	Performance	11.3.5	MIT	Open-source performance auditing tool for web pages	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Security Audit	Security Audit Framework	Active	Snyk - Open-source security scanner		Snyk	Security Audit	1.1000.0	Proprietary	Open-source security scanner for identifying vulnerabilities in code and dependencies	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Deployment Automation	Deployment Automation Framework	Active	Terraform - Infrastructure as Code		Terraform	Deployment Automation	1.10.0	Proprietary	Open-source infrastructure as code tool for managing cloud resources	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Monitoring Dashboard	Monitoring Dashboard Framework	Active	Grafana - Monitoring dashboard		Grafana	Monitoring Dashboard	10.3.3	Apache License	Open-source monitoring dashboard for visualizing metrics and logs	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Logging Aggregator	Logging Aggregator Framework	Active	Fluentd - Log aggregator		Fluentd	Logging Aggregator	1.15.0	Apache License	Open-source log aggregator for collecting and forwarding log data	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Queue Processor	Queue Processor Framework	Active	Celery - Distributed task queue		Celery	Queue Processor	5.4.0	MIT	Open-source distributed task queue for scheduling and executing tasks	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Cache Invalidation	Cache Invalidation Framework	Active	Memcached - Distributed memory cache		Memcached	Cache Invalidation	1.6.18	MIT	Open-source distributed memory cache for reducing database load	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Search Indexing	Search Indexing Framework	Active	Elasticsearch - Search engine		Elasticsearch	Search Indexing	8.14.0	Apache License	Open-source search engine for full-text and structured data	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Reporting Dashboard	Reporting Dashboard Framework	Active	Tableau - Business intelligence		Tableau	Reporting Dashboard	2023.3	Proprietary	Open-source business intelligence tool for data visualization and analysis	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
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Performance Framework	Performance Framework Framework	Active	Lighthouse - Performance auditing		Lighthouse	Performance Framework	11.3.5	MIT	Open-source performance auditing tool for web pages	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
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Monitoring Dashboard Framework	Monitoring Dashboard Framework Framework	Active	Grafana - Monitoring dashboard		Grafana	Monitoring Dashboard Framework	10.3.3	Apache License	Open-source monitoring dashboard for visualizing metrics and logs	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
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Security Audit Framework Framework	Security Audit Framework Framework Framework	Active	Snyk - Open-source security scanner		Snyk	Security Audit Framework Framework	1.1000.0	Proprietary	Open-source security scanner for identifying vulnerabilities in code and dependencies	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Deployment Automation Framework Framework	Deployment Automation Framework Framework Framework	Active	Terraform - Infrastructure as Code		Terraform	Deployment Automation Framework Framework	1.10.0	Proprietary	Open-source infrastructure as code tool for managing cloud resources	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Monitoring Dashboard Framework Framework	Monitoring Dashboard Framework Framework Framework	Active	Grafana - Monitoring dashboard		Grafana	Monitoring Dashboard Framework Framework	10.3.3	Apache License	Open-source monitoring dashboard for visualizing metrics and logs	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Logging Aggregator Framework Framework	Logging Aggregator Framework Framework Framework	Active	Fluentd - Log aggregator		Fluentd	Logging Aggregator Framework Framework	1.15.0	Apache License	Open-source log aggregator for collecting and forwarding log data	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Queue Processor Framework Framework	Queue Processor Framework Framework Framework	Active	Celery - Distributed task queue		Celery	Queue Processor Framework Framework	5.4.0	MIT	Open-source distributed task queue for scheduling and executing tasks	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Cache Invalidation Framework Framework	Cache Invalidation Framework Framework Framework	Active	Memcached - Distributed memory cache		Memcached	Cache Invalidation Framework Framework	1.6.18	MIT	Open-source distributed memory cache for reducing database load	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Search Indexing Framework Framework	Search Indexing Framework Framework Framework	Active	Elasticsearch - Search engine		Elasticsearch	Search Indexing Framework Framework	8.14.0	Apache License	Open-source search engine for full-text and structured data	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Reporting Dashboard Framework Framework	Reporting Dashboard Framework Framework Framework	Active	Tableau - Business intelligence		Tableau	Reporting Dashboard Framework Framework	2023.3	Proprietary	Open-source business intelligence tool for data visualization and analysis	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Compliance Framework Framework Framework	Compliance Framework Framework Framework Framework	Active	GDPR - General Data Protection Regulation		GDPR	Compliance Framework Framework Framework	1.0.0	Proprietary	Open-source framework for ensuring compliance with GDPR and other data privacy regulations	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Accessibility Framework Framework Framework	Accessibility Framework Framework Framework Framework	Active	WCAG - Web Content Accessibility Guidelines		WCAG	Accessibility Framework Framework Framework	2.1.4	Proprietary	Open-source guidelines for making web content more accessible to people with disabilities	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
Performance Framework Framework Framework	Performance Framework Framework Framework Framework	Active	Lighthouse - Performance auditing		Lighthouse	Performance Framework Framework Framework	11.3.5	MIT	Open-source performance auditing tool for web pages	Cloudinary	Development	1.33.0	Open-source cloud-based image and video management service	
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Last Name	First Name	Value		#DIV/	#DIV/		
Böhm	Luca			0!	0!		
Meusling	Patrick						
Arifin	Hafidz						
Huy	Christoph						
Selbig	Simon			0	No size		
Haseeb	Abdul			1	Trivial size		
Pohnke	Hannes			2	Small size		
Jokiel	Rene			3	Medium size		
Khabouze	Mehdi			5	Large size		
				8	Very large size		
				13	Too large (size)		
How to play planning poker							
1. Everyone type their number into their value field, don't hit return yet							
2. Someone, perhaps a product owner, count down 3.. 2.. 1..							
3. Then, everyone hit return to submit their value							